FLAIR Task Schematic

ICU Daily In-Hospital Mortality at 07:00 Local (Task T1)

FLAIR Benchmark

June 4, 2026

Abstract

We specify the FLAIR daily ICU mortality task: at every local 07:00 rounds anchor during the first ICU stay, predict whether the hospitalization will end with the patient dying in hospital (Expired or Hospice) rather than transferring to a Long-Term Acute Care Hospital (LTACH). The cohort is conditioned on non-survival — only hospitalizations whose eventual disposition maps to LTACH or in-hospital mortality are included — so the task is a binary terminal-trajectory question between two compatible end-of-life trajectories. Task T2 (LTACH discharge) is the exact label complement on the same cohort.

1 FLAIR operational definition

1.1 Anchors and windows

Let H_i denote a hospitalization with admission time t_i^{adm} and discharge or death time $t_i^{\text{end}} = \min(t_i^{\text{disch}}, t_i^{\text{death}})$. Let $t_i^{\text{ICU}_1,\text{start}}$ and $t_i^{\text{ICU}_1,\text{end}}$ be the first ICU stay endpoints returned by `compute_icu_timing` on the CLIF `ADT` table. All datetimes are naive but wall-clock-local per `cfg["timezone"]`; literal hours of the day are interpreted as local hospital time.

Prediction-window start (pred_start).

$$\text{pred_start}_i \;=\; \text{trunc}\big(t_i^{\text{ICU}_1,\text{start}},\, 1\text{d}\big) + 1\text{d} + 7\text{h}.$$

The first decision point is local 07:00 on the calendar day *after* first ICU admission. A patient admitted at 03:00 on Monday gets their first prediction at 07:00 on Tuesday.

Prediction-window stop (pred_stop).

$$\text{pred_stop}_i \;=\; \min\big(t_i^{\text{ICU}_1,\text{end}},\, t_i^{\text{disch}},\, t_i^{\text{death}}\big).$$

Predictions cease at the end of the first ICU stay or the end of the hospitalization, whichever comes first.

Decision-point grid.

Daily 24h step: $D_i = \{\text{pred_start}_i + k\cdot 24\text{h} : k \in \mathbb{Z}_{\geq 0},\, \text{pred_start}_i + k\cdot 24\text{h} < \text{pred_stop}_i\}$.

Label (`label_mortality`).

Let $\texttt{disp}_i = \texttt{lower}(\texttt{strip}(\texttt{discharge_category}_i))$. For $d \in D_i$:

$$y_i(d) \;=\; \mathbb{1}[\texttt{disp}_i \in \{\texttt{"expired"}, \texttt{"hospice"}\}]\,.$$

The label is constant across all rows of H_i (the hospitalization's eventual disposition is a single fact broadcast to every grid point).

1.2 Cohort (conditional on non-survival)

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- ≥ 1 ICU ADT record (first ICU stay is defined).
- First ICU stay long enough to contain at least one 07:00 grid point after the admission day (i.e., $\text{pred_start}_i < \text{pred_stop}_i$).
- `discharge_category` (case-insensitive) matches either:
  - `"expired"` or `"hospice"` (in-hospital mortality), or
  - regex `ltach|long.?term` (LTACH discharge).
- **Exclude** `discharge_category` null or any alive disposition (home, SNF, rehab, AMA, still admitted, other).

1.3 CLIF tables and categories used

Task T1 reads three CLIF v2.1 tables. Two carry a category-column filter that defines the cohort; the third is a join-only auxiliary for demographics and death time.

CLIF table	Category column	Literal values kept (case-insensitive)
`hospitalization`	`discharge_category`	{`"expired"`, `"hospice"`} ∪ regex `ltach|long.?term`
`adt`	`location_category`	{`"icu"`}
`patient`	—	join only; no category filter

Table 1: CLIF surface for Tasks T1 and T2. The `discharge_category` row is the *cohort* filter (LTACH or in-hospital mortality, applied before split). The T1 label additionally restricts to {`"expired"`, `"hospice"`} and the T2 label restricts to the LTACH regex; both labels are computed on the same retained set of rows.

Columns selected from each table:

- `hospitalization`: `hospitalization_id`, `patient_id`, `admission_dttm`, `discharge_dttm`, `age_at_admission`, `discharge_category`.
- `patient`: `patient_id`, `sex_category`, `race_category`, `ethnicity_category`, `death_dttm`.
- `adt`: `hospitalization_id`, `location_category`, `in_dttm`, `out_dttm`.

A FLAIR site whose local `discharge_category` vocabulary maps LTACH or long-term acute care under a different literal must extend the regex in `flair_benchmark/tasks/task1_icu_daily_mortality.py` (and Task T2 in lockstep) or the cohort will silently lose its LTACH-arm denominator.

2 Algorithmic flow

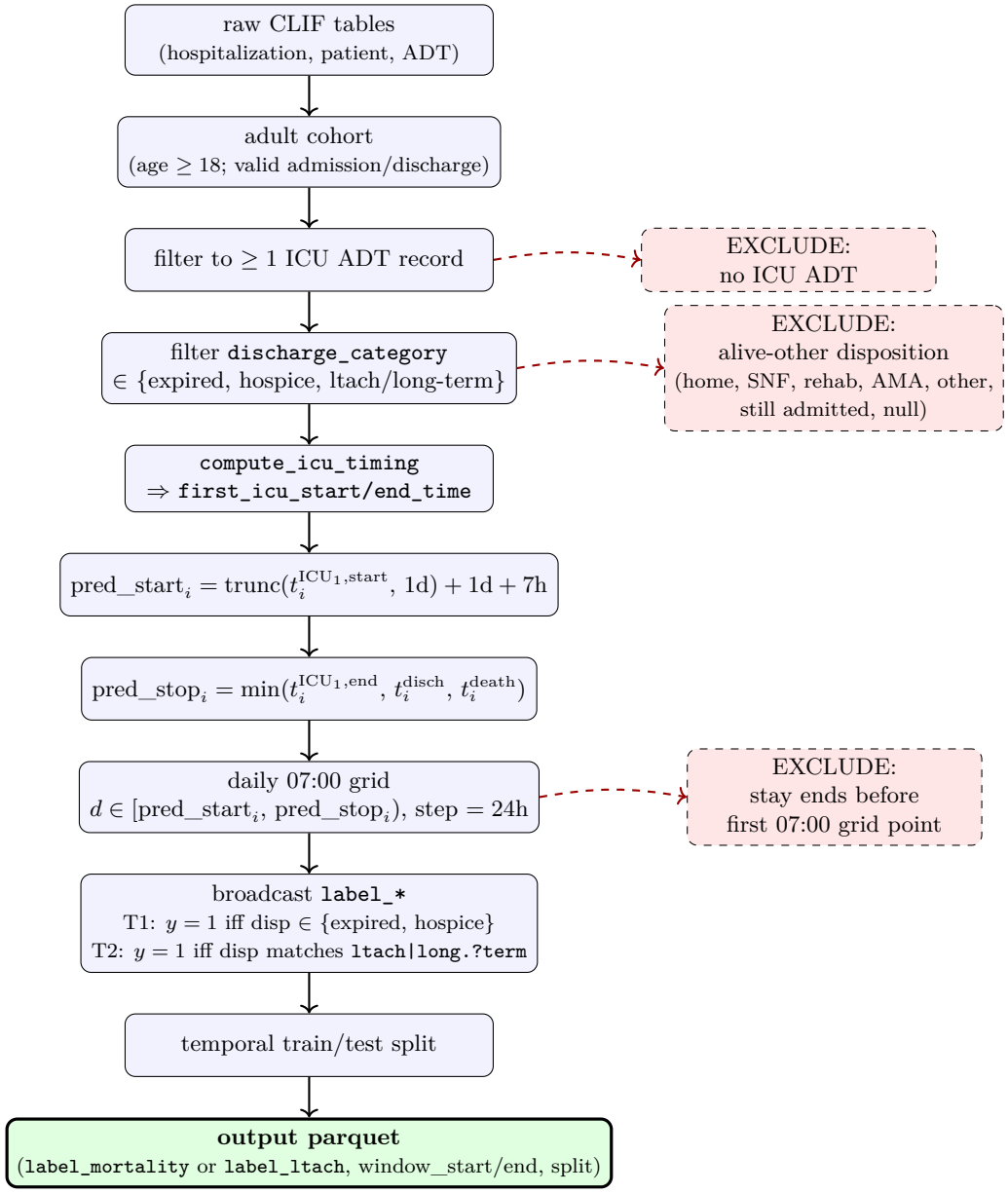


Figure 1: Build pipeline shared by Tasks T1 and T2. Solid arrows: main flow. Dashed red arrows: exclusions. The two tasks differ only in the final label box; both consume the same cohort, the same grid, and the same lookback window. Task T2 (LTACH) is detailed in a sibling schematic.

Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt`.
2. Build the adult cohort with valid admission/discharge times.
3. Restrict to hospitalizations with ≥ 1 ICU ADT row.
4. Filter `discharge_category` to LTACH OR in-hospital mortality (case-insensitive); drop nulls and alive-other dispositions.
5. Compute first-ICU-stay window via `compute_icu_timing(adt)`.
6. Compute pred_start$_i$ by truncating $t_i^{\text{ICU}_1,\text{start}}$ to date, adding 1 day, then adding 7 hours.
7. Compute pred_stop$_i = \min(t_i^{\text{ICU}_1,\text{end}}, t_i^{\text{disch}}, t_i^{\text{death}})$.
8. Explode the daily 07:00 grid [pred_start$_i$, pred_stop$_i$) per hospitalization (`pl.datetime_ranges`, `interval = "24h"`, `closed="left"`).
9. Broadcast the label (`disp`$_i \in$ {`"expired"`, `"hospice"`}) to every grid row.
10. Assign `window_start` $= d - 24$h, `window_end` $= d$ (fixed 24h lookback).
11. Apply temporal train/test split per `flair_benchmark._split.assign_split`, keyed on `prediction_dttm`.
12. Select `OUTPUT_COLS` and persist as parquet.

3 Design choices and rationale

Why a conditional-on-non-survival cohort. Among all ICU admissions the per-hospitalization mortality rate sits around 10–15%, so an unconditional mortality task is extremely imbalanced and dominated by trivial-negative examples (low-acuity elective admissions that go home). Conditioning on non-survival disposition (LTACH OR mortality) carves out the clinically meaningful fork: among patients whose eventual trajectory is one of two terminal arms, which one is it? This makes the binary task balanced enough to train without aggressive class-weighting and aligns with how the clinical question is asked at morning rounds for chronically critically ill patients.

Why 07:00 local rounds anchor. ICU morning rounds are the canonical decision cadence; predictions produced before rounds are actionable that day, predictions after rounds are not. The 07:00 timestamp is a wall-clock-local literal, relying on `flair_benchmark._clif.pd_to_pl` having stripped timezone info from `clifpy` datetimes (which themselves are typed to `cfg["timezone"]`). Across CLIF sites this is a stable choice; sites that prefer 06:00 or 08:00 can re-anchor by adjusting `PRED_HOUR` without changing the rest of the pipeline.

Why daily (not hourly). The decision being modeled — terminal-trajectory commitment — does not change hour-to-hour. A daily cadence matches the clinical question and keeps row counts manageable (MIMIC-CLIF: 12,438 stays / 65,989 daily rows; see `docs/paper/outline.tex` Table 1).

Prevalence and the asymmetry with Task T2. At the row level the positive rate is 57.4% (mortality), with stays-ever-positive 67.1%. The conditional cohort is NOT 50/50 between mortality and LTACH; in MIMIC, mortality dominates because hospice and expired-in-hospital both fall on the mortality side of the fork while only LTACH carries the alternative. Task T2 (LTACH) ends up with a complementary but asymmetric positive rate of 42.6% per-row (32.9% stays-ever-positive).

4 Out of scope

- Multi-class extension to alive-other dispositions (home, SNF, rehab, AMA) as a third class.
- Hospital-wide (non-ICU) variant.
- Input feature engineering and severity-score baselines (e.g., APACHE-III (Knaus et al., 1991), SAPS-II).
- Survival-time / time-to-event variant.

References

William A. Knaus, Douglas P. Wagner, Elizabeth A. Draper, Jack E. Zimmerman, Marilyn Bergner, Paulo G. Bastos, Carl A. Sirio, Donald J. Murphy, Ted Lotring, Anne Damiano, and Frank E. Harrell Jr. The APACHE III prognostic system: Risk prediction of hospital mortality for critically ill hospitalized adults. *Chest*, 100(6):1619–1636, 1991. doi: 10.1378/chest.100.6.1619.